

CSA1711 - Artificial Intelligence

CO2 Assessment Tool - 2

MCQ Test - Questions, Options and Answer Key

Q1. In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. This approach focuses on selecting the node that appears closest to the goal without considering the cost already incurred.

Which algorithm follows this principle?

- A) Breadth First Search
- B) Uniform Cost Search
- C) Greedy Best-First Search ✓ Correct Answer**
- D) Depth First Search

Justification: Selects nodes purely by estimated closeness to the goal (heuristic $h(n)$), ignoring accumulated cost.

Q2. A* search is widely used in pathfinding and graph traversal because it combines the actual cost from the start node and the estimated cost to the goal. This balance allows it to find optimal solutions efficiently when the heuristic is admissible.

What is the evaluation function used in A* search?

- A) $f(n) = g(n)$
- B) $f(n) = h(n)$
- C) $f(n) = g(n) + h(n)$ ✓ Correct Answer**
- D) $f(n) = g(n) \times h(n)$

Justification: A* combines actual cost so far $g(n)$ with estimated remaining cost $h(n)$: $f(n) = g(n) + h(n)$.

Q3. Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. A good heuristic should not overestimate the actual cost and should improve search efficiency.

What property ensures optimality in A* search?

- A) Completeness
- B) Admissibility ✓ Correct Answer**
- C) Depth limit
- D) Backtracking

Justification: An admissible heuristic never overestimates true cost, guaranteeing A* finds the optimal path.

Q4. Local search algorithms operate by starting with an initial solution and iteratively improving it based on a given objective function. These methods are commonly used in optimization problems where the path is not important but the final solution is.

Which of the following is a local search algorithm?

- A) A* Search
- B) Hill Climbing ✓ Correct Answer**
- C) Breadth First Search
- D) Uniform Cost Search

Justification: Hill Climbing starts from an initial state and moves to better neighbouring states; the path taken is irrelevant.

Q5. In adversarial search, agents compete against each other in a game environment. The decision-making process involves considering both the agent's moves and the opponent's possible responses.

Which algorithm is commonly used for such decision-making?

- A) Greedy Search
- B) Minimax Algorithm ✓ Correct Answer**
- C) Backtracking
- D) Uniform Search

Justification: Minimax models optimal play for both the MAX and MIN players by evaluating a game tree.

Q6. A delivery robot in a warehouse selects paths based only on the estimated distance to the destination without considering obstacles already encountered. This sometimes leads to suboptimal paths but faster decisions.

Which algorithm is the robot using?

- A) A* Search
- B) Greedy Best-First Search ✓ Correct Answer**
- C) Uniform Cost Search
- D) Depth Limited Search

Justification: This is the defining behaviour of Greedy Best-First Search - heuristic only, no path cost considered.

Q7. A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination. This helps in finding the shortest path efficiently.

Which algorithm best represents this approach?

- A) Depth First Search
- B) A* Search ✓ Correct Answer**
- C) Hill Climbing
- D) Backtracking

Justification: The $g(n) + h(n)$ combination (distance travelled + estimated remaining distance) is characteristic of A* Search.

Q8. A university is scheduling exams for multiple subjects with constraints such as no overlapping exams and limited rooms. The solution must satisfy all constraints while assigning time slots.

This problem is best modeled as:

- A) Game Playing Problem
- B) Constraint Satisfaction Problem ✓ Correct Answer**
- C) Heuristic Search Problem
- D) Optimization Problem

Justification: Variables (exams), domains (time slots/rooms), and constraints (no overlap) define a Constraint Satisfaction Problem.

Q9. An AI-based chess program evaluates all possible moves and anticipates the opponent's responses before making a decision. It assigns utility values to game states to determine the best move.

Which algorithm is being applied here?

- A) Greedy Search
- B) Hill Climbing

C) Minimax Algorithm ✓ *Correct Answer*

D) Breadth First Search

Justification: This describes the Minimax algorithm's alternating MAX/MIN evaluation of game states using utility values.

Q10. A robot explores an unknown environment where it does not have prior knowledge of obstacles. It makes decisions step-by-step and updates its knowledge as it moves.

Which type of agent is suitable for this scenario?

A) Offline Agent

B) Online Search Agent ✓ *Correct Answer*

C) Static Agent

D) Deterministic Agent

Justification: An Online Search Agent interleaves sensing, acting, and planning in real time as it explores an unknown environment.

Q11. A* search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. If these conditions are violated, the algorithm may still find a solution but not necessarily the best one.

Why is an admissible heuristic important?

A) It reduces memory usage

B) It ensures optimality ✓ *Correct Answer*

C) It increases branching factor

D) It avoids loops

Justification: An admissible (never-overestimating) heuristic is what guarantees A* returns the optimal solution.

Q12. In large game trees, evaluating every node becomes computationally expensive. Alpha-Beta pruning helps eliminate branches that do not affect the final decision, thereby improving efficiency.

What is the primary benefit of Alpha-Beta pruning?

A) Reduces depth of tree

B) Reduces number of nodes evaluated ✓ *Correct Answer*

C) Changes game rules

D) Improves heuristic accuracy

Justification: It prunes branches that cannot influence the final decision, reducing the number of nodes evaluated.

Q13. Hill Climbing algorithm continuously moves toward better states based on local improvements. However, it may fail to reach the global optimum due to certain limitations in the search space.

What is the main drawback of Hill Climbing?

A) High memory usage

B) Revisiting nodes

C) Getting stuck in local maxima ✓ *Correct Answer*

D) Requires complete tree

Justification: Hill Climbing can settle on a state better than its neighbours but worse than the global optimum (local maxima).

Q14. Constraint Satisfaction Problems involve variables, domains, and constraints. A valid solution must assign values to all variables such that no constraints are violated.

What ensures a valid CSP solution?

- A) Heuristic function
- B) Path cost
- C) Constraint satisfaction ✓ Correct Answer**
- D) Depth-first traversal

Justification: A valid CSP solution is an assignment of values to all variables that satisfies every constraint.

Q15. In adversarial games, evaluation functions are used when it is not feasible to explore the entire game tree. These functions estimate the desirability of a game state.

What is the role of an evaluation function?

- A) Generate successors
- B) Estimate utility of a state ✓ Correct Answer**
- C) Reduce branching factor
- D) Store moves

Justification: It estimates the desirability/utility of a non-terminal game state as a substitute for deep search.

Q16. In A* search, the evaluation function is calculated using both the path cost and heuristic value. Suppose a node has a path cost of 7 and a heuristic estimate of 5.

What is the value of $f(n)$?

- A) 12 ✓ Correct Answer**
- B) 2
- C) 35
- D) 7

Justification: $f(n) = g(n) + h(n) = 7 + 5 = 12$.

Q17. A heuristic function sometimes overestimates the actual cost to reach the goal. This violates the admissibility condition required for optimal solutions in A* search.

What happens in such a case?

- A) A* remains optimal
- B) A* becomes non-optimal ✓ Correct Answer**
- C) A* becomes BFS
- D) A* stops working

Justification: Without admissibility, A* is no longer guaranteed to find the optimal path - it becomes non-optimal.

Q18. Consider a search tree with a branching factor of 2 and depth of 4. The total number of nodes generated in a full binary tree can be calculated analytically.

What is the approximate number of nodes?

- A) 10
- B) 15
- C) 31 ✓ Correct Answer**
- D) 16

Justification: Sum over levels 0-4: $2^0 + 2^1 + 2^2 + 2^3 + 2^4 = 1 + 2 + 4 + 8 + 16 = 31$.

Q19. In a minimax tree, each node has 3 children and the depth of the tree is 3. The total number of leaf nodes depends on the branching factor raised to the depth.

How many leaf nodes are present?

- A) 9
- B) 27 ✓ Correct Answer**
- C) 6
- D) 12

Justification: Leaf nodes = $b^d = 3^3 = 27$.

Q20. Alpha-Beta pruning improves the efficiency of minimax by reducing the number of nodes evaluated. In the best-case scenario, it significantly lowers time complexity.

What is the best-case time complexity?

- A) $O(b^d)$
- B) $O(b^{(d/2)})$ ✓ Correct Answer**
- C) $O(d^b)$
- D) $O(\log b)$

Justification: With optimal move ordering, Alpha-Beta reduces complexity from $O(b^d)$ to $O(b^{(d/2)})$.
